Pandelis Zembashis

CTO | Staff Applied AI Engineer | Agent Architecture & Developer Tools

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EXECUTIVE SUMMARY

CTO and hands-on applied AI engineer building agent systems for software development. At Cosine, built and led much of the runtime behind AI coding agents: context handling, tool use, multi-agent orchestration, deterministic replay, repository search, browser use, memory, evaluation loops, and secure execution across CLI, desktop, VS Code/WebAssembly, cloud, and enterprise deployments. Strong fit for Bolt.new through deep agent architecture, TypeScript/product engineering, multi-provider LLM work, eval-driven iteration, and a developer-tooling background focused on helping users go from intent to working software.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – May 2026

AI coding lab building agents, developer tools, model feedback loops, and enterprise deployment infrastructure.

- Led product and engineering for Cosine's AI coding platform, setting technical direction across agent architecture, tool use, context management, runtime reliability, developer experience, and enterprise deployment.
- Built the core Go agent harness for streaming tool calls, event scheduling, checkpoints, deterministic replay, memory writes, MCP/LSP integrations, subagents, file locks, repository search, browser use, and structured rollout traces.
- Designed shared agent execution semantics across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, keeping model/tool behavior consistent across local, browser-adjacent, and cloud-backed surfaces.
- Built evaluation and debugging loops from real coding sessions: replayable traces, test results, LSP diagnostics, checkpoints, reviews, conversation patterns, and parallel sandboxed runs to improve agent behavior systematically.
- Worked across OpenAI, Anthropic, Gemini, and internal model capabilities, shaping prompts, tool contracts, context strategy, and provider-specific behavior into product-level patterns.
- Built secure remote execution for model-controlled coding work: Kubernetes workers, gVisor/runsc isolation, ephemeral sessions, network boundaries, and private VPC, on-prem, and air-gapped deployment paths.
- Helped position Cosine as a full-stack AI lab with benchmark-leading coding models, a production coding-agent platform, and public recognition for sovereign AI infrastructure.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled LLM-backed outbound automation systems that processed millions of email chains and billions of tokens, with event-driven architecture, worker orchestration, and operational monitoring.
- Established engineering cadence, onboarding, incident response habits, and cross-functional execution across product, engineering, and GTM during rapid growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a production fleet management platform in Next.js, React Query, Tailwind CSS, and TypeScript.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving iteration across web, operations, and marketplace workflows.
- Established development cadences, product intake, and stakeholder feedback loops that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

SELECTED OSS, WRITING & DEVELOPER TOOLING

OSS: codex-web-sdk; claude-agent-sdk-rust; claude-agent-sdk-go; gut; jalla; 100+ merged public GitHub PRs.
Cosine Agent Systems: Why we built our own harness; Cosine Swarm; Secure AI coding agents for enterprise.
Developer Tooling: Cosine CLI 1.16: built-in search, single-binary distribution, OS pipes, Go/Rust integration, and ripgrep-level internal search architecture.
Talks & Public Material: AWS Startups - The Future Of Work; Directus TV - Edge Computing; pa.ndel.is; Devpost portfolio.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

Agent Architecture: Context management, tool use, multi-step workflows, subagents, MCP/LSP, browser use, memory, deterministic replay
Applied LLM Systems: OpenAI, Anthropic, Gemini, prompt systems, provider strategy, evaluations, traces, model/tool behavior analysis
Developer Tools & Product Engineering: TypeScript, Node.js, React, Next.js, GraphQL, CLI tooling, IDE integrations, WebAssembly, SDK design
Platform & Reliability: Go, Kubernetes, gVisor, sandboxes, observability, event logs, checkpoints, private deployment
Leadership: Staff technical direction, cross-team roadmaps, mentoring, product strategy, technical writing, high-growth teams